

Improved Delayed Neutron Group Parameters for Molten Salt Reactors

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Outline

① Introduction

Defining Terms

② Motivation

Cat Behavior

Cat Appearance

Cat Math

③ Methods

Illinois Colors

④ Conclusion



What are Delayed Neutrons?

Delayed neutron precursors (DNPs) are nuclides that undergo β^- decay, producing a daughter, called an emitter, which emits a delayed neutron [2].

Relevant data for DNPs includes:

$P_{n,i}$ = the emission probability [-]

τ_i = the half-life of DNP i [s]

$\chi_{d,i}$ = the delayed neutron energy spectrum from DNP i [-]

N_i = the number of atoms of i [-].

Accurately modeling the DNPs is important for transient simulations.



What are DNP groups and parameters?

A small number of artificial DNPs, known as DNP groups, are used to approximate the DNPs to reduce computational cost.

The three parameters for DNP groups are:

$\bar{\nu}_{d,k}$ = the delayed neutron yield of DNP group k [-]

τ_k = the half-life of DNP group k [s]

$\chi_{d,k}$ = the delayed neutron energy spectrum of DNP group k [-]

Six of these DNP groups can accurately approximate the DNPs in a light water reactor [1].



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An Image



Figure 1: A caption describing the image. [?].

A Table

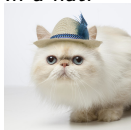
Frames (slides) can have “blocks.”

This one is about a cat

A cat in a hat.

A cat

In a hat.



Columns



Sometimes things need to be put side by side, in two nice looking columns. Maybe one column involves a quotation.

Explicit is better than implicit. – The Zen of Python

And, also, perhaps, a logo.



Figure 2: A caption describing the image.
[?].

Some Code

I have to use the fragile syntax for code slides.

```
def meow(volume):  
    """Make a demanding noise at the specified volume  
  
    Parameters  
    -----  
  
    volume: int  
        The volume of the demand. No relation to importance.  
  
    Returns  
    -----  
  
    str  
        meow  
    """  
  
    o = 'o'*volume  
    return 'me'+ o + 'ow'
```

Cat Math: Part 1



$$x = y \tag{1}$$

where

$$x = \text{cats} \tag{2}$$

$$y = \text{peculiar} \tag{3}$$

Cat Math: Part 2

Everything in Beamer is just like in $\text{\LaTeX}$. Right down to the theorems.

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$





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Blue and Orange are Fierce

Those are the Illini Colors. Use them like you see them in Figure 3.



Figure 3: Kristofer Hivju is pretty serious about this color palette [?].

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Conclusion

We showed many things. This slide is an example of how you can animate bulleted lists, for more information about using beamer animations, checkout the overleaf article on overlay specifications in the group's guide.

- Cats are peculiar

Conclusion



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- Cats are peculiar
- Blue and Orange are fierce colors
- Math can be rendered nicely

Conclusion

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- Cats are peculiar
- Blue and Orange are fierce colors
- Math can be rendered nicely
- Cite your sources

Acknowledgement

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If you are funded by an NEUP grant, that number usually goes here. .

References I

[1] MC Brady and TR England.

Delayed neutron data and group parameters for 43 fissioning systems.

Nuclear Science and Engineering, 103(2):129–149, 1989.

[2] NEA Data Bank.

The JEF-2.2 nuclear data library. JEFF report 17, 2000.